12.0 Maintenance and PM Schedule

Maintenance Task	Frequency	Recommendations
Inspect auger for stretch	Quarterly	If auger has stretched passed the designated distance from the end cap it will need to be trimmed. Refer to Appendix A for instructions.
Inspect casing	Quarterly	Look for any warping, holes, or areas where the casing maybe thinning out. Also inspect for any residual material build up.
Inspect center core	Quarterly	Look for any signs of damage or wear. Replace if needed.
Inspect compression coupling gaskets	Quarterly	Identify if gaskets are worn, warped, torn, or missing. If so replace.
Clean residual material out of hopper	Daily/Weekly	Helix hoppers should have residual material removed from the hopper if the unit is going to be inactive for more than 4 hours.
Inspect hopper boot end cap for damage or wear	Quarterly	Remove the end cap and inspect for any wear, damage or gasket degradation. Replace if needed.
Check oil level in gear box	Quarterly	Refer to your Hapman job manual for the specific fill level for your gear box.

12.0 Maintenance and PM Schedule

Maintenance Task	Frequency	Recommendations
Inspect drive discharge box top cover gaskets	Quarterly or at any auger change over	Identify if gaskets are worn, warped, torn, or missing. If so replace.
Inspect U-bolt assembly for wear or damage	Quarterly or at any auger change over.	Verify Nyloc nuts are intact, the clamp plate is not bent, and the U-bolt is not damaged.
Document running amperages with and without material	Quarterly	Keep a running log of historic amperages. *Reference pages 65-66 for instructions.*
Verify E-stop, safety sensor*, or mechanical interlock *are working properly.	Weekly	Press E-stop and verify loss of control power. Disengage safety sensor or mechanical interlock and verify loss of control power.

13.0 Troubleshooting

Issue	Checks
Helix will not start	1. Check safety relay and verify no E-stops are pressed and that all safety sensors are active 2. Verify mechanical interlock is engaged and working* 3. Verify VFD does not have an active fault code 4. Confirm that the 24vdc power supply is on and outputting 24vdc 5. Check contact relays and confirm they are coming on when told
Helix stops running unexpectedly	1. Check that the low-level sensor is active, and the output light is on. 2. Verify on the face of the VFD that it doesn't have an active fault.
Casing is packing with material	1. Verify that the conveyor discharge does not have material build up or blockage present. 2. If equipped, remove the center core from the unit and run. 3. If using a beveled edge auger confirm that it is installed correctly as called out on *pages 80-81*.
Auger is breaking	1. Verify that auger is 2" from the end cap as called out on *page 44*. 2. Check to see if material is packing in the casing or at the discharge. 3. Inspect to see if any residual material has solidified in the casing that would prevent or impede the rotation of the auger. 4. Make sure material being ran matches the description that was provided to Hapman. 5. Contact Hapman Customer Service to further investigate.

13.0 Troubleshooting

Issue	Checks
Helix is running but little to no material is coming out of the conveyor discharge.	1. Check to see if the material in the infeed hopper is bridging or rat holing. 2. Inspect to see that Helix auger is intact and undamaged.
Helix conveyor is running too slow or too fast.	1. Using the vendor manual adjust the run speed of the VFD. 2. Contact Hapman Customer Service
NOTICE	***If after performing the checks the problem persists, please contact Hapman Customer Service.***

****Not all Helix conveyors have mechanical interlocks. While the terminals are present in all Helix™ panels, we place a jumper around the interlock from the factory if a safety interlock device is being utilized instead. Reference your electrical drawings or contact Hapman Customer Service with any questions.***

14.0 Recommended Mechanical Spare Parts

Qty	Description
1	Helix Auger
1	Helix Casing
1	Helix Center Core*
2	Compression Couplings
2	U-bolt Assembly
1	Gear Reducer

NOTICE

Non-factory approved parts can cause improper operation and could cause unsafe operating conditions.

****Not all Helix conveyor applications require a center core. Reference your mechanical drawings or contact Hapman Customer Service with any questions.***

15.0 Recommended Electrical Spare Parts

Qty	Description
1	Low-Level Sensor
1	Low-Level Sensor Cable
1	VFD*
2	Control Relays
1	Safety Interlock Sensor*

NOTICE

Non-factory approved parts can cause improper operation and could cause unsafe operating conditions.

****Not all Helix control panels contain VFD’s or safety interlock sensors. Reference your electrical drawings or contact Hapman Customer Service with any questions.***

16.0 Hapman Customer Service

For assistance with questions regarding installation, operation, maintenance, warranty, or equipment troubleshooting, please contact Hapman Customer Service.

E-mail: service@hapman.com
Toll Free: 1-800-427-6260
U.S.A. Phone: 269-343-1675

Hapman Customer Service Parts Group

For assistance with ordering parts, tracking parts orders, or information about parts lead times or pricing please contact the Hapman Customer Service Parts Department.

E-mail: parts@hapman.com
Toll Free: 1-800-427-6260
U.S.A. Phone: 269-343-1675

Hapman Equipment Information

Hapman Job #________________________________

Date of Installation______________________________